

I am writing to express my interest in the Research Assistant position on lattice Boltzmann methods for wind energy applications at the University of Oldenburg.

EDUCATION

I chose to study aerospace engineering. It took me a few years, and a detour through wind farms and black holes, to find my way back. The following subsections trace that path and the research direction it has led to.

Undergraduate Studies

Studying aerospace engineering at TU Delft allowed me to understand the art of simplifying complex phenomena to solve engineering challenges, yet I found myself drawn to acquire knowledge in a field of science where I could explore the art of precision and delve further into the *why* question. My constant curiosity and drive have led me into the cycle of gaining and passing down knowledge within the scientific community, where I strive to collaborate and truly make an impact. To move closer to the source of the methods I was using, I sought out research at the interface of mathematics and engineering. In a project with [Bart Ripperda](#) and [Sean Ressler](#), through numerical relativistic fluid simulations of Rayleigh–Taylor instabilities, we demonstrated that a dense outer accretion disk penetrates an inner dilute magnetosphere, facilitating gas particle movement to the event horizon. This instilled in me a resolute desire to study advanced mathematics and modern physics, which led me to a minor in astrophysics at the University of British Columbia.

In the last two years of my undergraduate studies, I worked with [Marc Gerritsma](#) on bundle-valued forms in mimetic spectral element methods, and it was in this project that I first met analysis, PDEs and numerical methods as subjects in their own right rather than as auxiliary tools. The project introduced me to the idea that a discretisation should not only approximate a PDE, but also retain its structural features at the discrete level. This was my first serious encounter with the analytic and geometric questions underlying numerical methods. Motivated by this, I followed the Applied Mathematics bridging programme at TU Delft in my third year, alongside my regular aerospace coursework and my BSc thesis on regenerative wind farming (supervised by [Carlos Simão Ferreira](#) and [Uwe Fechner](#)), to prepare for graduate study in mathematics.

Graduate Studies

This path led me to the MSc in Mathematical Modelling and Scientific Computing at the University of Oxford. Guided by my academic advisor, [Endre Süli](#), I completed a few supererogatory courses on functional analysis and functional analytic methods for PDEs. These courses granted me the privilege of conducting a dissertation on [energy- and enstrophy-preserving discretisations of the incompressible Navier Stokes equations](#) under the supervision of [Patrick Farrell](#). Beyond the dissertation, I implemented physics-informed neural networks in PyTorch with [Kathryn Gillow](#), and studied high-dimensional Lévy flights for novelty search with [Peter Grindrod](#).

2026 (Gap year)

Alongside the more abstract work, I have wanted at least part of my mathematics to begin at a table, with children and a piece of chalk. That wish followed me home when I left Oxford and returned to Iran, where I came across a rural village whose only school survived as a building and a list of names, with no teacher left to open the door. For a short time I became the missing teacher, opening the door each morning, writing on a borrowed board, and showing a handful of children how to line numbers up on a page and make sense of them. Since then, with the help of local supporters and investors, we have raised enough in funding and commitments to repair the school and guarantee a proper salary for a permanent teacher. With the school now open, my role has become that of someone they can call on. I can only ask to devote years to the mathematics I love if, somewhere at the edge of the map, there are children sitting in a classroom with at least the chance to love it too.

I then co-founded *trew* with [Pablo Cembellín](#), whom I met at Oxford, and served as its CTO. The idea was to give firms tamper-evident, cryptographically verifiable proof of what their AI systems actually decided, a requirement that regulators are increasingly imposing. Advised by [Luca Enriques](#), we took the product to a complete MVP. We did not close a funding round, and are now seeking an acquirer with the help of [Ben Marzouk](#).

I am currently translating [Medieval Finitism](#) by [Mohammad Saleh Zarepour](#) into Farsi, with the help of [Kaveh Lajevardi](#). The book examines medieval arguments against actual infinity and traces their role in the later mathematical treatment of infinities of different sizes.

--- PUBLICATIONS & PREPRINTS

My dissertation has since been extended into a paper, currently under review at *Foundations of Computational Mathematics*.¹ In this work, we have proposed and analysed a mixed finite element discretisation for the incompressible Navier–Stokes equations that preserves the evolution of both energy $\mathcal{K}$ and enstrophy $\mathcal{E}$ at the discrete level. In two dimensions, the boundedness of enstrophy yields a Reynolds-robust bound on $\|\nabla \mathbf{u}\|$, providing a natural stabilisation mechanism for under-resolved flows. Conforming implementations rest on discrete Stokes complexes with $\mathbf{H}(\text{grad curl})$ – or H^2 –conforming vorticity spaces, which are available only in limited polynomial degrees and mesh configurations. We have therefore introduced two reformulations: a symmetric interior penalty formulation over the standard discrete de Rham complex, and an $\mathbf{H}(\text{grad curl})$ – or H^2 –free reparametrisation via auxiliary variables using only standard grad–, curl– and div–conforming spaces. A broad range of boundary conditions can be incorporated. Numerical experiments confirm the stabilising effect of enstrophy preservation in both 2D and 3D.

Seminars & Conferences

Elements of this work have been presented at a number of international conferences and workshops. I presented it in a minisymposium talk at [Firedrake’25](#) (University of Leeds) and in a poster at [SIAM UKIE ’26](#) (University of Leicester). [Boris Andrews](#) presented it at [ECCOMAS / MFET](#) (RWTH Aachen University), [ACOMEN](#) (Ghent University), the [ESI Programme on Differential Complexes](#) (University of Vienna), [SciCADE](#) (University of Edinburgh), a workshop on finite element tensor calculus (Tsinghua University), and the Numerical Analysis Group seminar (University of Oxford).

--- WHY THIS PROJECT

Wind energy is where my path back to mathematics began: my BSc thesis on regenerative wind farming was my first sustained encounter with the flows this project targets. My research since has centred on discretisations of the Navier–Stokes equations that stay stable for under-resolved, high-Reynolds-number flows, precisely the regime of atmospheric turbulence and turbine wakes. Lattice Boltzmann methods attack the same problem from the kinetic side, and the question of which structure a scheme preserves, and how that governs its stability, carries over directly. Together with my experience of large-scale simulation codes such as [BHAC](#) and [Athena++](#), and of machine-learned turbulence models in [OpenFOAM](#), this project would let me return to wind energy with the numerical tools I have built since.

¹Boris D. Andrews, Matin Shams, and Patrick E. Farrell. *Strongly enstrophy-stable integrators for the incompressible Navier-Stokes equations*. 2026. DOI: [10.48550/arXiv.2609.15520](#). arXiv: 2609.15520.